applied optics

On-chip multifunctional polarizer based on phase change material

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Received 14 August 2023; revised 28 September 2023; accepted 29 September 2023; posted 29 September 2023; published 13 October 2023

Polarizers are used to eliminate the undesired polarization state and maintain the other one. The phase change material $Ge_2Sb_2Se_4Te_1$ (GSST) has been widely studied for providing reconfigurable function in optical systems. In this paper, based on a silicon waveguide embedded with a GSST, which is able to absorb light by taking advantage of the relatively large imaginary part of its refractive index in the crystalline state, a multifunctional polarizer with transverse electric (TE) and transverse magnetic (TM) passages has been designed. The interconversion between the two types of polarizers relies only on the state switching of GSST. The size of the device is 7.5 µm ∗ 4.3 µm, and the simulation results showed that the extinction ratio of the TE-pass polarizer is 45.37 dB and the insertion loss is 1.10 dB at the wavelength of 1550 nm, while the extinction ratio (ER) of the TM-pass polarizer is 20.09 dB and the insertion loss (IL) is 1.35 dB. For the TE-pass polarizer, a bandwidth broader than 200 nm is achieved with ER > 20 dB and IL < 2.0 dB over the wavelength region from 1450 to 1650 nm and for the TM-pass polarizer, ER > 15 dB and IL < 1.5 dB in the wavelength region from 1525 to 1600 nm, with a bandwidth of approximately 75 nm.

https://doi.org/10.1364/AO.503268

1. INTRODUCTION

Controlling light at the chip scale has been a topic of long-standing interest in integrated photonics. Silicon on insulator (SOI) [1] photonics waveguides show great potential due to their ability to allow sharp bending of the waveguide and their ability to be densely integrated on a chip. Most of the devices in silicon photonics circuits are sensitive to the polarization state of the input light, so on-chip polarization processing devices are necessary because of the need for precise control of TE and TM modes. Polarization-related properties in integrated optics have attracted much attention in recent years. SOI has become one of the most studied device platforms due to its high refractive index contrast caused by its geometrical anisotropy [2], which is intrinsically highly polarization dependent. Polarization processing devices such as polarizers [3,4], polarization rotators [5,6], and polarization beam splitters [7,8] have become the most typical applications of SOI platforms [1,2,9]. If a polarization division multiplex is not necessary, a common approach consists of using a polarizer to strip off the unwanted polarization [10,11].

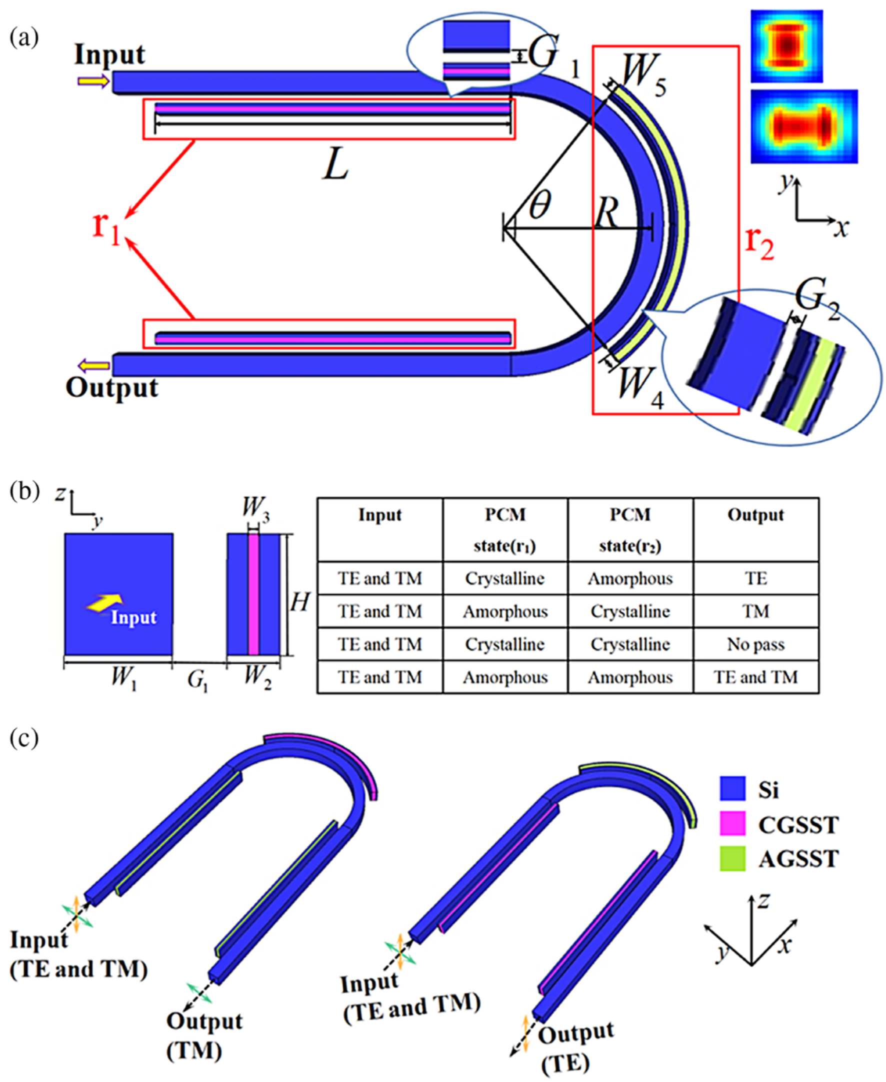
Typically, a polarizer is a filter that passes light of a certain polarization direction and blocks polarized light in the orthogonal direction. In an optical polarizer, the loss of one polarization state is greater than the loss of another polarization state. As light travels a distance within the device, the two orthogonally polarization states form a polarization extinction ratio (ER), which is the ratio of energy between the passing and non-passing polarization state [12]. In general, optical polarizers based on different structures can be grouped according to their working mechanism: polarizers based on directional couplers [13], polarizers based on subwavelength gratings [12,14], polarizers based on hybrid plasmonic waveguides [15,16], and so on. The authors have proposed a photo-switch polarizer based on azobenzene dye. By remotely re-aligning the azobenzene dye molecules into lateral or vertical direction using a polarizing blue light, one can accordingly control the TE- or TM-passed polarizer [17]. However, all reported Si waveguide-based polarizers are TE-pass or TM-pass polarizers. For integrated optical systems that require polarization management, polarizers with reconfigurable features can be considered in operating environments where multifunctional polarization is required [18]. To achieve active control, one option is to mix structures with new materials. For example, graphene is one of the useful materials for constructing broadband polarizers, which enables significant changes in the optical constants of the waveguide, and graphene attenuates the TM polarization state and TE polarization state differently. However, some graphene-based polarizers have been shown to be inappropriate for using isotropic models when dealing with the TM polarization state. In addition, the insertion loss (IL) of these structures is still quite high due to scattering in gratings and strong optical absorption by graphene/metal [19,20]. Another option is to mix waveguides with phase change materials (PCMs), which have been widely studied for their ability to provide active tuning/switching in optical systems [13,21,22]. Among PCMs, $Ge_2Sb_2Te_5$ (GST) [23] has superior optical properties to the well-known material vanadium dioxide [24]. Reversible phase transitions between the amorphous and crystallin states of GST can be achieved in subnanoseconds. The phase change of the GST can be thermally, optically, or electrically induced [23]. Its optical properties, such as refractive index and absorption, show considerable variation between the amorphous and crystalline states. In addition, the switching state is self-maintained at each material stage without the power consumption of a continuous state. In recent years, very small GST elements (10 nm thick) loaded on silicon nitride nanophotonic waveguides have been used for the development of compact integrated optical non-volatile memories [25], suggesting that GST is a candidate material for the realization of active control of multifunctional optical components. Despite these attractive features, GST still suffers from excessive optical losses, even in the amorphous state, due to its small bandgap and the resulting interband absorption in the telecommunication bands. This fundamentally limits many applications. A new, low-loss optical phase change material has recently been reported that can overcome the limitations of a GST-based optical switching device. $Ge_2Sb_2Se_4Te_1$ (GSST) [21,22], derived in GST by partially substituting Te with Se, increases the optical bandgap, enabling reduced loss in the 1310 nm and 1550 nm telecommunication bands and relatively low phase change temperature. Similar to GST, GSST material can be triggered either electrically (through thermal conduction heating or electrical threshold switching) or optically. So, different phase states of GSST material could be achieved during the practical device fabrication process. The crystallization ratio of GSST material can be controlled by the width, power, and irradiation time of the laser pulse. In particular, the imaginary part of GSST is much smaller than that of GST and thus shows considerable promise to be integrated with a SOI platform for the design of low-loss, non-volatile switching photonic devices. The refractive index of AGSST has a relatively small real part (~3), whereas the refractive index of CGSST has a relatively large real part (~5), and the imaginary part of their refractive indices differs a lot [26]. The imaginary part of the crystalline state is larger, which has the effect of absorbing light under certain circumstances. So in this paper, GSST is embedded in the silicon waveguide for coupling [18] and eliminating the unwanted polarization states in the U-shaped silicon waveguide, thus leaving the desired polarization states for polarization.

In this paper, we propose a multifunctional polarizer based on a hybrid GSST–Si waveguide. The key effective blocks are formed by two polarizer regions which are able to manage the TE and TM modes, respectively. For the first region (r_1), two hybrid GSST–Si straight waveguides are introduced to impact the TM mode inputted. For the second region (r_2), the bend GSST–Si waveguide is used to affect the TE mode inputted. Whether these two regions will take effect depends on the phase states of the GSST layer which are embedded in the silicon chambers. From results, we have obtained four output cases (TE, TM, None, TE + TM) for the input fundamental TE and TM modes through such a device only by changing the phase states of the embedded GSST layers within a relative compact footprint (<8 μm). The simulation results predicted 1.35 dB IL and 20.09 dB ER for the TM-pass polarizer. The TE-pass polarizer exhibits an IL of 1.10 dB and an ER of 45.37 dB at 1550 nm. We hope the present scheme can provide a useful way to construct multifunctional photonic integrated circuits.

2. DEVICE STRUCTURE AND PRINCIPLE

Figure 1(a) is the top view. The entire device has a U-shaped structure. It consists of two straight waveguides and one semicircular waveguide. The two curved waveguides in the top view are part of two concentric circles, so the coupling spacing at the curved waveguides is equal everywhere. Figure 1(b) is the cross-sectional view of the device. The cross-sectional view shows two regular rectangles with the PCMs interspersed in the center of the waveguide, and the height of the waveguide is $H = 340$ nm. The polarizer is symmetric up and down and is surrounded by silicon dioxide. At a wavelength of 1550 nm, the material indices are $n_{Si} = 3.45$ and $n_{SiO_2} = 1.44$. As shown in Fig. 1(a), since the TE polarization state is constrained laterally and the TM polarization state is constrained vertically, in a rectangular waveguide, both polarization states move toward the outer portion of the rectangular waveguide in a sufficiently small bending process, whereas the TE polarization state moves more in comparison to the TM polarization state and the smaller the radius of the bending, the more displacements are generated, which makes the bending waveguide favorable to be used for coupling the TE polarization state as well [27]. Therefore, we used a bending waveguide to couple the TE polarization state to obtain a TM-pass polarizer. This is done by embedding AGSST at r_1 and CGSST at r_2, as shown in Fig. 1(c), and pattern-matching the bending Si waveguide with r_2 based on the TE mode, so that the TE mode can be coupled and absorbed to obtain a TM-pass polarizer. Similarly, CGSST is embedded at r_1 and AGSST is embedded at r_2, and the straight Si waveguide is pattern-matched with r_1 based on the TM mode, so that the TM mode can be coupled and absorbed to obtain a TE-pass polarizer. In addition, since the coupling entrance is only 300 nm, but we put a trapezoidal Si waveguide at the input port to make an articulation, one side of the trapezoid is 500 nm long, and the other side is 300 nm long, and after simulation, we found that the light transitions from the width of 500 nm to 300 nm, the loss is very small, and the transmittance reaches 0.999. The other performances of the polarizer are also not significantly affected.

As mentioned before, the imaginary part of the CGSST is relatively large and can absorb light, so when it is necessary to couple and eliminate a certain polarization state, the CGSST is embedded in the corresponding silicon waveguide, and the waveguide coupling length, coupling spacing, and the waveguide width are adjusted by watching the performance of the device through pattern matching and simulation, at which time the AGSST has almost no effect on the light. Therefore, there



Input	PCM state(r_1)	PCM state(r_2)	Output
TE and TM	Crystalline	Amorphous	TE
TE and TM	Amorphous	Crystalline	TM
TE and TM	Crystalline	Crystalline	No pass
TE and TM	Amorphous	Amorphous	TE and TM

Fig. 1. (a) Top view and (b) cross section of the polarizer. (c) Three-dimensional schematic of the on-chip multifunctional polarizer based on PCMs. Material and structural parameters are labeled.

are four cases here. First, when the CGSST is embedded in the straight ridge waveguides (r_1) and mode matching is done based on the TM polarization state, and the AGSST is embedded in the curved ridge waveguide (r_2), when the light propagates in the U-shaped waveguide, the TM polarization state will be coupled away and absorbed at the straight waveguide, and the final output is the TE polarization state, as shown in Fig. 1(c). Second, when the AGSST is embedded in the straight ridge waveguides (r_1) and the CGSST is embedded in the curved ridge waveguide (r_2) and mode matching is done based on the TE polarization state, when the light propagates in the U-shaped waveguide, the TE polarization state will be coupled away and eliminated at the curved waveguide, and the final output will be the TM polarization state, as shown in Fig. 1(c). Third, when a CGSST is embedded in both the straight ridge waveguides (r_1) and the curved ridge waveguide (r_2), and mode matching is done based on the TM polarization state at the straight waveguide, while mode matching is done based on the TE polarization state at the bent waveguide, the light propagating through the U-shaped waveguide then couples away and absorbs the TM polarization state and TE polarization state at the straight and bent waveguides, respectively, and ends up with no output. Fourth, when an AGSST is embedded in both the straight ridge waveguides (r_1) and the curved ridge waveguide (r_2), when light propagates in the U-shaped waveguide, there is very little coupling and absorption of light at both the straight and bent waveguides, so that finally both TM and TE will output. The above four scenarios are visually summarized in the functional table of Fig. 1.

3. RESULTS AND DISCUSSION

The device parameters are designed and optimized using the three-dimensional finite-difference time-domain method (3D-FDTD) [28] for the device performance analysis. A good polarizer should have a high polarization ER, low IL, and small device size [19,20]. For the TM-pass polarizer, the mode ER and IL are defined as [29]

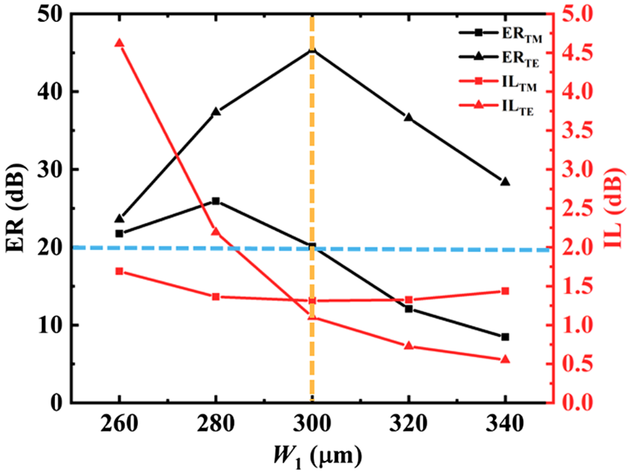


Fig. 2. ER and IL of the TE-pass polarizer and TM-pass polarizer as a function of waveguide width and $\lambda = 1550$ nm.

$$\mathrm{ER_{TM}} = 10\log_{10}(T_{\mathrm{TM}}/T_{\mathrm{TE}}), \quad (1)$$

$$\mathrm{IL_{TM}} = -10\log_{10}(T_{\mathrm{TM}}). \quad (2)$$

For the TE-pass polarizer, the mode ER and IL are defined as

$$\mathrm{ER_{TE}} = 10\log_{10}(T_{\mathrm{TE}}/T_{\mathrm{TM}}), \quad (3)$$

$$\mathrm{IL_{TE}} = -10\log_{10}(T_{\mathrm{TE}}), \quad (4)$$

where P_{TM} and P_{TE} stand for the receiving power of the TM mode and TE mode at the device output port, respectively. If not otherwise specified, the working wavelength is set as 1550 nm in the following discussion and we fix the waveguide widths W_2 as 147 nm and W_4 as 193 nm. The bending radius is then determined to be $R = 2$ μm considering the relatively large bending loss and the current fabrication feasibility. After extensive numerical simulations and comparative analyses are carried out, the optimized values are as follows: the embedded block widths $W_3 = 31$ nm and $W_5 = 108$ nm, the waveguide width $W_1 = 300$ nm, the GSST length in r_1 $L = 5$ μm, the gap between the assisted and main waveguide $G1 = 150$ nm and $G_2 = 153$ nm, and the angle of the circle center in r_2 $\theta = 104°$.

First, in order to determine the width of the waveguide W_1, the ER and IL of the TE-pass polarizer and TM-pass polarizer, respectively, were simulated as a function of the waveguide width in the FDTD. As shown in Fig. 2, in order to make the ER > 20 dB, the waveguide width is as wide as possible, so W_1 is determined as 300 nm.

In order to determine the waveguide coupling length and coupling spacing at the straight waveguide, the transmittances at the output of TE-pass polarizer and TM-pass polarizer are simulated in FDTD as shown in Figs. 3(a)–3(d), respectively, and the corresponding ERs and ILs are calculated based on these transmittances. The ERs and ILs are plotted as a function of the waveguide coupling length and coupling spacing to obtain the optimal waveguide coupling length and coupling spacing. As shown in Fig. 4(a), with the change of coupling length at the straight waveguide (r_1), the ER and IL of the TE-pass polarizer are greatly affected. The ERs of

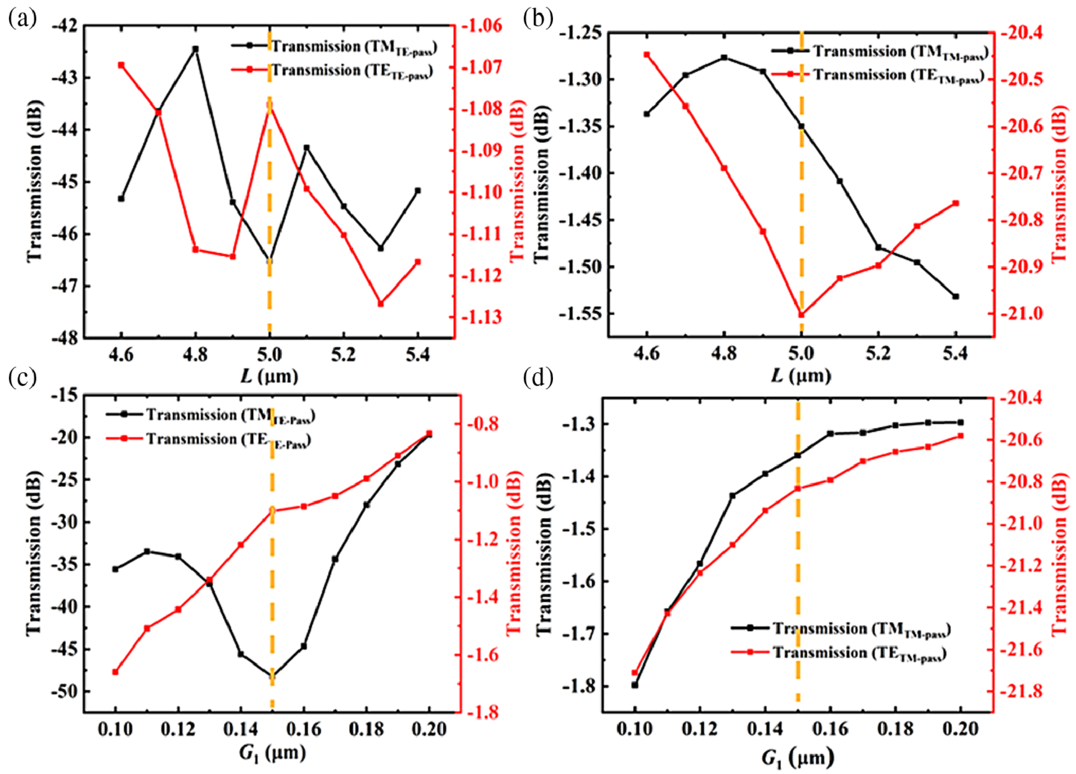


Fig. 3. L-dependent transmittances of the (a) TE-pass polarizer and (b) TM-pass polarizer; G_1-dependent transmittances of the (c) TE-pass polarizer and (d) TM-pass polarizer.

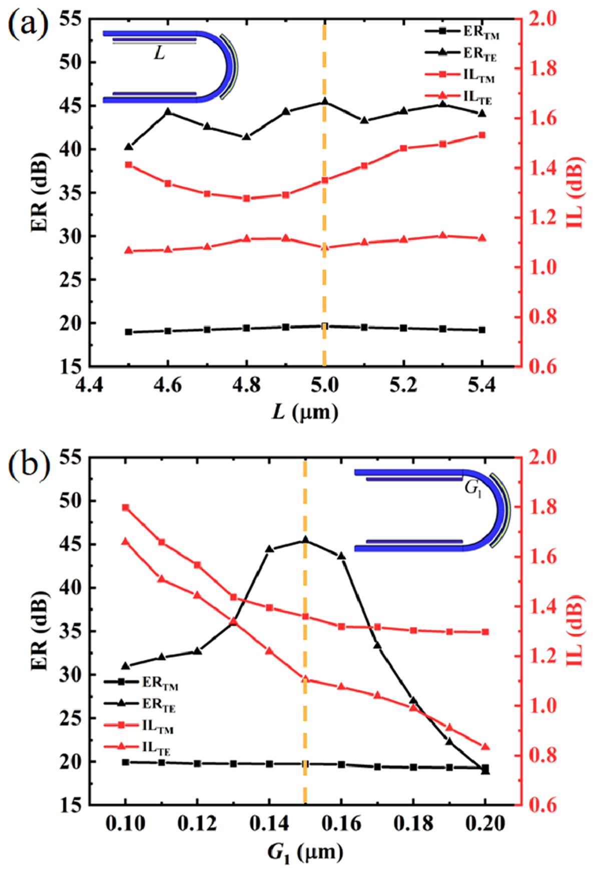


Fig. 4. ER and IL of the TE-pass polarizer and TM-pass polarizer as a function of (a) coupling length and (b) coupling spacing at the straight waveguide and $\lambda = 1550$ nm.

the TE-pass polarizer and TM-pass polarizer exhibit high value ($ER_{TM} > 19$ dB and $ER_{TE} > 40$ dB) and low loss ($IL_{TE} < 1.2$ dB and $IL_{TM} < 1.5$ dB) within the calculation range from 4.5 to 5.3 μm. Such large dimensions greatly broaden the fabrication tolerance. In order to make the ER as large as possible and the IL as small as possible, the coupling length is taken to be 5 μm. As shown in Fig. 4(b), with the change of coupling spacing at the straight waveguide, the ER and IL of the TE-pass polarizer are also greatly affected; in order to take a compromise value between the ER and IL, the coupling spacing is chosen to be 150 nm.

The transmittances at the output of TE-pass polarizer and TMpass polarizer are simulated in FDTD as shown in Figs. 5(a)–5(d), respectively, and the corresponding ERs and ILs are calculated based on these transmittances. The ERs and ILs are plotted as a function of the waveguide coupling length and coupling spacing to obtain the optimal waveguide coupling length and coupling spacing. Figures 6(a) and 6(b) show the performance of the device with respect to the coupling length θ and coupling spacing G_2 at the bent waveguide (r_2), and the simulations are carried out in FDTD to obtain the optimal waveguide coupling length and coupling spacing, where L and G_1 are set as their optimum values aforementioned. As shown in Fig. 6(a), the IL of the TM-pass polarizer exhibits a gradual increase as the coupling length at the curved waveguide increases, while the IL of the TE-pass polarizer exhibits a gradual decrease. The value corresponding to this parameter has the standard of $ER_{TM} > 20$ dB and $IL_{TM} < 1.5$ dB in the interval [104°,114°] and as shown in Fig. 6(b), when G_2 is larger than 153 nm, the coupling to the TE mode is weakened due to the coupling spacing at the bent waveguide becoming larger. The value corresponding to this parameter has the standard of $ER_{TM} > 20$ dB

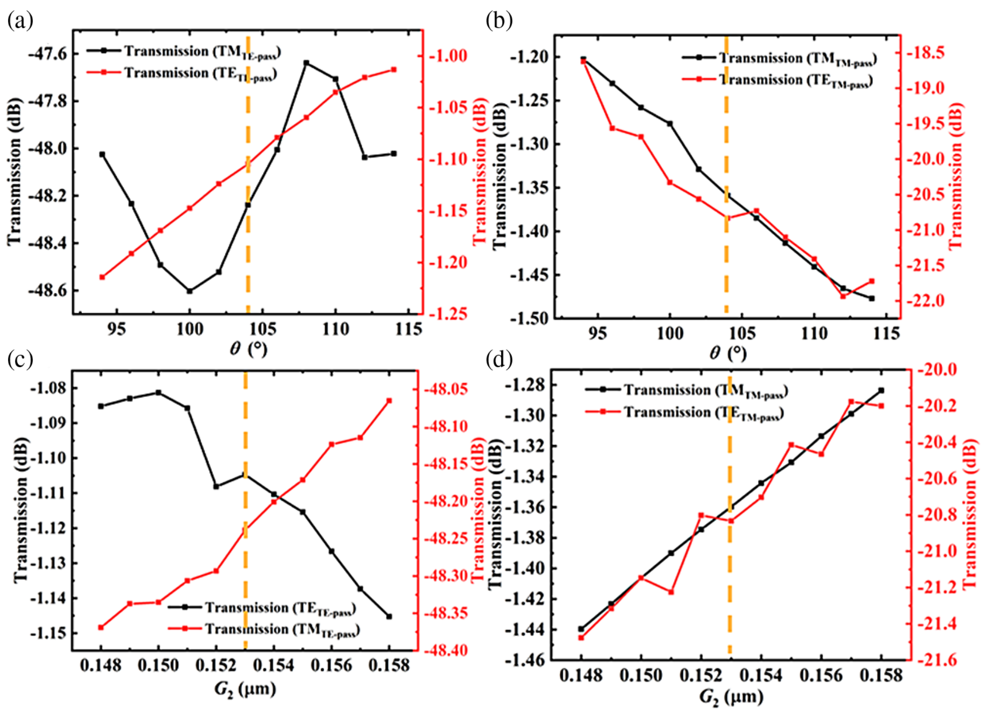


Fig. 5. θ-dependent transmittances of the (a) TE-pass polarizer and (b) TM-pass polarizer; G_2-dependent transmittances of the (c) TE-pass polarizer and (d) TM-pass polarizer.

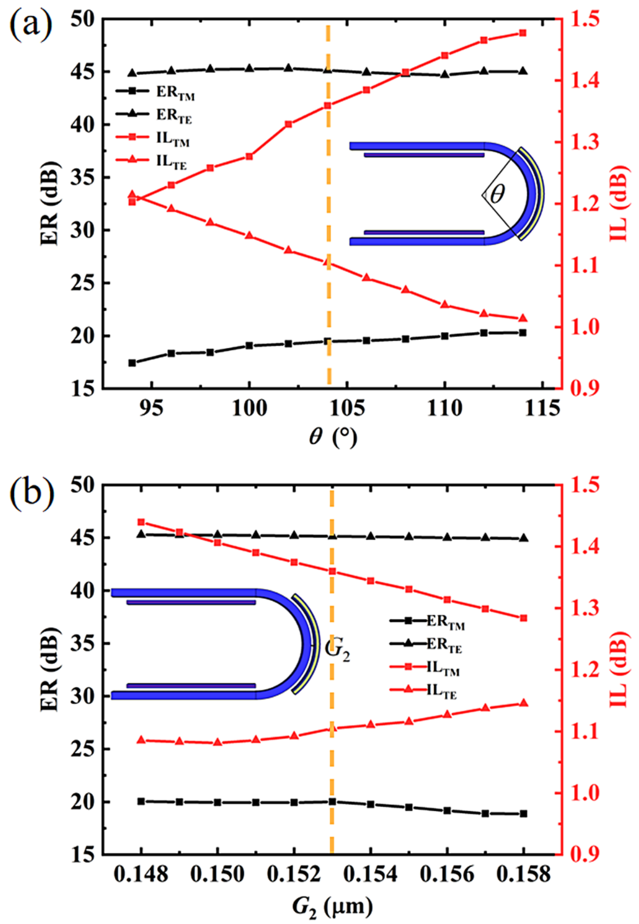


Fig. 6. ER and IL of the TE-pass polarizer and TM-pass polarizer as a function of (a) coupling length and (b) coupling spacing at the bent waveguide and $\lambda = 1550$ nm.

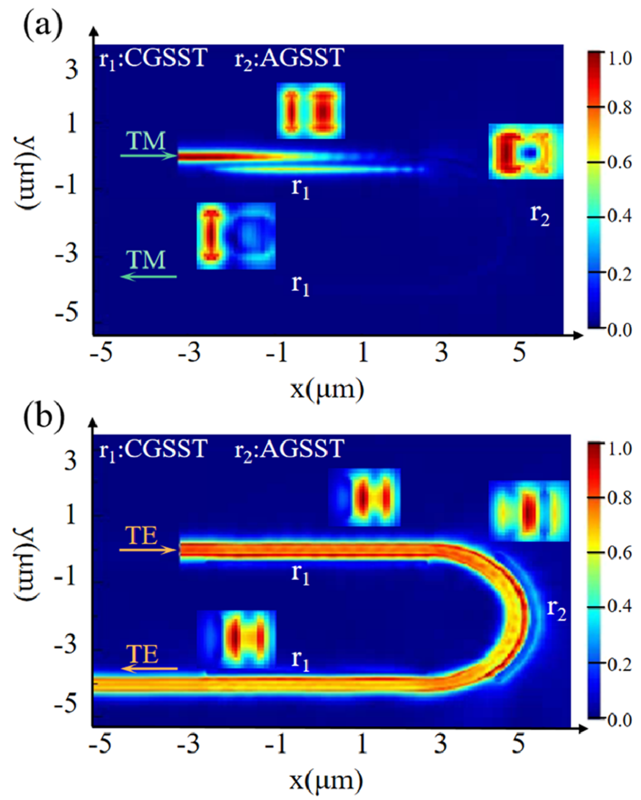


Fig. 7. (a) and (b) are the XY plane power propagation maps of the proposed polarizer of TM and TE polarization states at $\lambda = 1550$ nm; r_1 is CGSST and r_2 is AGSST.

and $IL_{TM} < 1.45$ dB in the interval [148 nm, 153 nm]. The interval value ranges above mentioned relax the requirements of the fabrication tolerance in the actual manufacturing process. By weighing the performance and compactness of the device, the optimum coupling length and coupling spacing are found at $\theta = 104^\circ$ and $G_2 = 153$ nm, respectively, and the width of the GSST layer is chosen as $W_5 = 108$ nm.

The dielectric constant ε_{eff} of GSST material is closely related to the material crystallization ratio η, which is defined as

$$\frac{\varepsilon_{\text{eff}}(\lambda) - 1}{\varepsilon_{\text{eff}}(\lambda) + 2} = \eta \times \frac{\varepsilon_c(\lambda) - 1}{\varepsilon_c(\lambda) + 2} + (1 - \eta) \times \frac{\varepsilon_a(\lambda) - 1}{\varepsilon_a(\lambda) + 2}, \quad \textbf{(5)}$$

where ε_c is the dielectric constant of GSST material in a completely crystalline state, and ε_a is the dielectric constant of GSST material in a completely amorphous state.

When the embedded at the straight waveguide (r_1) is a CGSST and the embedded at the bent waveguide (r_2) is an AGSST, and the mode matching at the straight waveguide is based on the TM polarization state, in this way, when the device is in operation, the TM polarization state is highly coupled and impacted during the propagation process, while the TE polarization state is very little affected, thus only one polarization can be transmitted out. Figures 7(a) and 7(b) simulate the energy propagation of TM and TE polarization states in the TE-pass polarizer. The TM polarization state is clearly coupled into the adjacent hybrid waveguide during propagation and is gradually absorbed. Since the TM polarization state is obstructed a lot, there is very little TM polarization state left at the output of the waveguide, which makes the ER of the TE-pass polarizer large. The FDTD simulation results showed that the ER of the TE-pass polarizer is 45.37 dB and the IL is 1.10 dB.

On the contrary, as shown in Fig. 8, when the AGSST is embedded at the straight waveguide and the CGSST is embedded at the curved waveguide, and the mode matching is based on the TE polarization state at the curved waveguide, in this way, when the device is in operation, the TE polarization state is highly attenuated in the propagation process, while the TM polarization state is very little affected. It can also be visualized from Fig. 8(a) that the TE polarization state has almost no loss at the straight waveguide, while it is suddenly coupled and absorbed by the GSST–Si hybrid waveguide next to the U-shaped waveguide when it propagates to the curved waveguide. In contrast, Fig. 8(b) shows that the TM-polarized state is almost unaffected throughout the propagation process. This shows that the design of the device in this paper is feasible. The FDTD simulation results show that the ER of the TM-pass polarizer is 20.09 dB, and the IL is 1.35 dB.

From the above, it can be seen that the designed polarizer in this paper has tunability and can be switched between the TE-pass polarizer and TM-pass polarizer only by adjusting the states (AGSST and CGSST) of GSST in the GSST–Si hybrid waveguide at both r_1 and r_2. In addition, this polarizer has two other functions, i.e., neither the TE nor the TM mode can be passed and both the TE and the TM modes can be passed, and

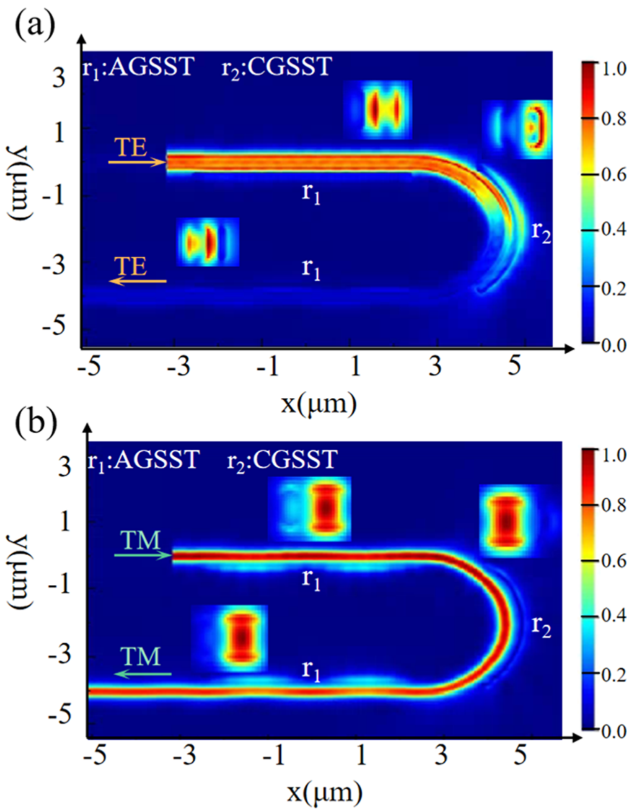


Fig. 8. (a) and (b) are the XY plane power propagation maps of the proposed polarizer of TE and TM polarization states at $\lambda = 1550$ nm; r_1 is AGSST and r_2 is CGSST.

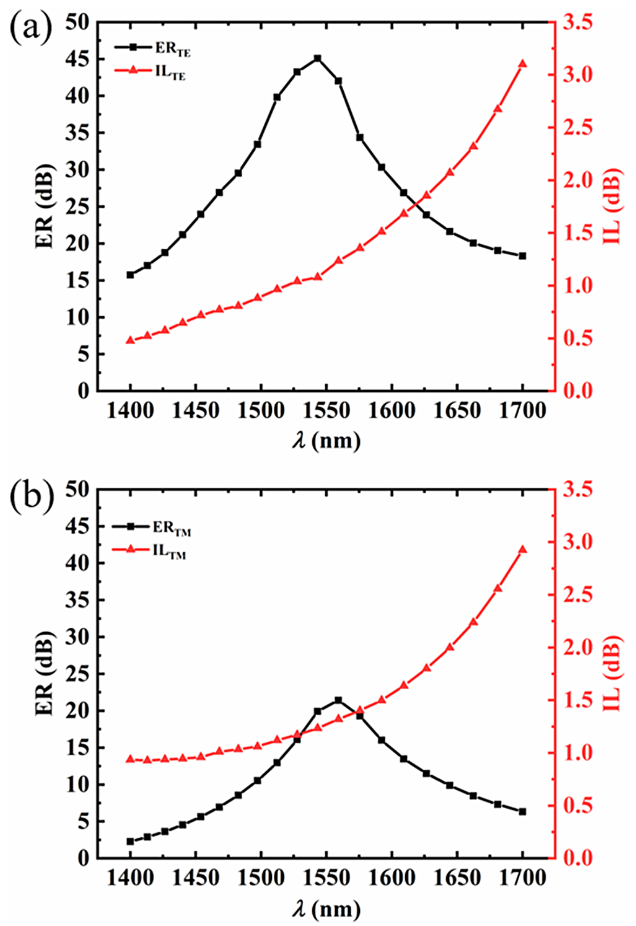


Fig. 9. Wavelength spectra of IL and ER for (a) TE-pass polarizers and (b) TM-pass polarizers. The waveguide range is calculated from 1400 to 1700 nm.

the ER of both the TE-pass polarizer and the TM-pass polarizer is >20 dB, and the IL is <1.5 dB, which makes the performance of the device pretty good, while the functions are varied.

Figure 9 depicts the ER and IL as a function of the wavelength spectrum of the device and considers both the TE-pass polarizer and the TM-pass polarizer cases, respectively. As a multifunctional device, for the TE-pass polarizer the corresponding ER and IL are >20 dB and <2.0 dB, respectively, over the calculated wavelength range of 1450–1650 nm. For the TM-pass polarizer, the corresponding ER and IL are >15 dB and <1.5 dB, respectively, over the calculated wavelength range of 1525–1600 nm. This exhibits a small relative wavelength dependence. Therefore, the proposed multifunctional polarizer has a large allowable operating bandwidth and good performance.

A multifunctional polarizer is designed based on the above dimensional parameters and state switching of PCMs, which can do the four choices for the following cases by changing the crystalline and amorphous states of the GSST. The crystallization ratio of the GSST material can be controlled by the width, power, and irradiation time of the laser pulse. First, when the CGSST is embedded in the straight ridge waveguides (r_1) and mode matching is done based on the TM polarization state, and the AGSST is embedded in the curved ridge waveguide (r_2), when the light propagates in the U-shaped waveguide, the TM polarization state will be coupled away and absorbed at the straight waveguide, and the final output is the TE polarization state, as shown in Fig. 1(c). Second, when the AGSST is embedded in the straight ridge waveguides (r_1) and the CGSST is embedded in the curved ridge waveguide (r_2) and mode matching is done based on the TE polarization state, when the light propagates in the U-shaped waveguide, the TE polarization state will be coupled away and eliminated at the curved waveguide, and the final output will be the TM polarization state, as shown in Fig. 1(c). Third, when a CGSST is embedded in both the straight ridge waveguides (r_1) and the curved ridge waveguide (r_2), and mode matching is done based on the TM polarization state at the straight waveguide, while mode matching is done based on the TE polarization state at the bent waveguide, the light propagating through the U-shaped waveguide then couples away and absorbs the TM polarization state and TE polarization state at the straight and bent waveguides, respectively, and ends up with no output. Fourth, when an AGSST is embedded in both the straight ridge waveguides (r_1) and curved ridge waveguide (r_2), when light propagates in the U-shaped waveguide, there is very little coupling and absorption of light at both the straight and bent waveguides, so that finally both TM and TE will output.

To better show the structural parameters of the final optimized device, a detailed table is presented as shown in Table 1, and Table 2 compares the performance of the devices designed in this paper with some previously reported polarizers. It can

Table 1. Device Parameters of the Proposed On-Chip Multifunctional Polarizer Based on Phase Change Material

Parameters	L (μm)	H (nm)	G_1 (nm)	G_2 (nm)	θ (°)	R (μm)	W_1 (nm)	W_2 (nm)	W_3 (nm)	W_4 (nm)	W_5 (nm)
Value	5	340	150	153	104	2	300	147	31	193	108

Table 2. Comparison of Several Polarizers

Device	Type	Type of Work	Length (μm)	ER (dB)	IL (dB)	Bandwidth (nm)
[1]	TE/TM	simulation	1.0	∼15	∼3	60 (IL < 3 dB)
[30]	TM pass	simulation	23.0	20	2.5	140 (ER > 20 dB, IL < 2.5 dB)
[31]	TM pass	simulation	5.2	20	0.32	155 (ER > 20 dB, IL < 0.32 dB)
[32]	TE/TM	simulation	1.0	∼18	2.6/0.8	100 (IL_{TM} < 0.8 dB, IL_{TE} < 2.6 dB)
[33]	TE pass	simulation	29.39	29.8	1.04	>80 (IL < 1.04 dB)
[34]	TE/TM	simulation	5.95/1.92	23/19	2.2/1.94	200 (ER > 19 dB)
This work	TE/TM	simulation	7.5	47.13/20.09	1.10/1.35	200/75 (ER_{TE} > 20 dB, IL_{TE} < 2 dB; ER_{TM} > 15 dB, IL_{TM} < 1.5 dB)

be observed that the performance of the device in this paper is not worse than other devices, while its biggest advantage is its versatility and tunability.

4. CONCLUSION

In conclusion, this paper demonstrates a multifunctional polarizer based on a GSST–Si hybrid waveguide structure. The proposed TE-pass and TM-pass polarizers show polarization tunability by exploiting the oversized variation of refractive index in GSST between the crystalline and amorphous states. The device size is 7.5 μm ∗ 4.3 μm, and the ER_{TE} is 45.37 dB and IL_{TE} is 1.10 dB at 1550 nm for the polarizer with TE passage, while the ER_{TM} is 20.09 dB and IL_{TM} is 1.35 dB for the polarizer with TM passage. Simulation results also showed that the TE-pass polarizer operates with a bandwidth of 200 nm in the range of 1450 to 1650 nm and the TM-pass polarizer operates with a bandwidth of 75 nm in the range of 1525 to 1600 nm. The developed GSST–Si hybrid method is very useful for multifunctional polarized optical devices for numerous applications and is a very promising option.

Funding. National Natural Science Foundation of China (62205129); Natural Science Foundation of Jiangsu Province (BK20200592); Fundamental Research Funds for the Central Universities (JUSRP12024); Graduate Research and Innovation Projects of Jiangsu Province (SJCX23_1223).

Disclosures. The authors declare no conflicts of interest.

Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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